

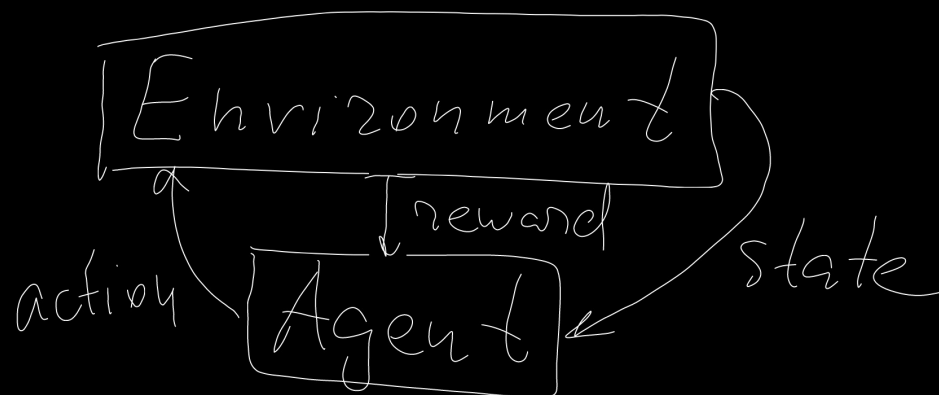
ML / DL

$$\{x_i, y_i\}_{i=1}^N, \quad y \approx f(x, \theta)$$

$$\frac{1}{N} \sum_{i=1}^N L(y_i, f(x_i, \theta)) + R(\theta) \rightarrow \min_{\theta}$$

↑
differentiable loss

RL



$s \in S$ - state

$a \in A$ - action

$\pi(a|s)$ -
- policy

$p(s'|s, a)$ -
- transition prob

$r(s, a)$ - reward

$$\tau := \{s_0, a_0, r_0, s_1, a_1, r_1, \dots\}$$

Markov Decision Process (MDP)

$$p(s_{t+1} | s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0) = \\ = p(s_{t+1} | s_t, a_t)$$

Cummulative reward:

$$R_0 = r_0 + r_1 + \dots + r_{T_{\max}}$$

Discounted cum. reward:

$$R_0 = r_0 + \gamma r_1 + \gamma^2 r_2 + \dots = \sum_{t=0}^{+\infty} \gamma^t r_t, \quad \gamma \in [0, 1)$$

$$\pi(a|s, \theta)$$

$$\mathbb{E}_{p(\tau|\theta)} R_0 \rightarrow \max_{\theta}$$

$$p(\tau|\theta) = p(s_0) \pi(a_0|s_0, \theta) p(s_1|s_0, a_0) \times \pi(a_1|s_1, \theta) \dots$$

Cross-Entropy Method (CEM)

$$x \sim p(x), \quad s(x) \in \mathbb{R} \qquad q_{\text{opt}}(x) = \frac{1}{Z} [s(x) \geq \tau] p(x)$$

$$\mathbb{E}_{p(x)} [s(x) \geq \tau] \quad ?$$

$$Z = \int [s(x) \geq \tau] p(x) dx$$

$$\begin{aligned} \mathbb{E}_{p(x)} [s(x) \geq \tau] &= \int [s(x) \geq \tau] p(x) dx = \\ &= \int [s(x) \geq \tau] \frac{p(x)}{q(x)} q(x) dx \approx \frac{1}{M} \sum_{j=1}^M \frac{[s(x_j) \geq \tau] p(x_j)}{q(x_j)} \Big|_{x_j \sim q(x)} \end{aligned}$$

$$q(x|\lambda), \quad KL(q_{opt}(x) \parallel q(x|\lambda)) \rightarrow \min_{\lambda}$$

$$KL = \mathbb{E}_{q_{opt}} \log \frac{q_{opt}(x)}{q(x|\lambda)} = \text{const} - \mathbb{E}_{q_{opt}} \log q(x|\lambda) \rightarrow \max_{\lambda}$$

$$\mathbb{E}_{q_{opt}} \log q(x|\lambda) = \frac{1}{Z} \mathbb{E}_{p(x)} [s(x) \geq \tau] \log q(x|\lambda) \rightarrow \max_{\lambda}$$

$$\mathbb{E}_{p(x)} [s(x) \geq \tau] \log q(x|\lambda) \approx \frac{1}{M} \sum_{j=1}^M \frac{[s(x_j) \geq \tau] p(x_j)}{q(x_j|\lambda_{old})} \log q(x_j|\lambda) \rightarrow \max_{\lambda}$$

$x_j \sim q(x|\lambda_{old})$

CEM scheme

Init. $M, p \in (0, 1), \lambda_0, t = 0$

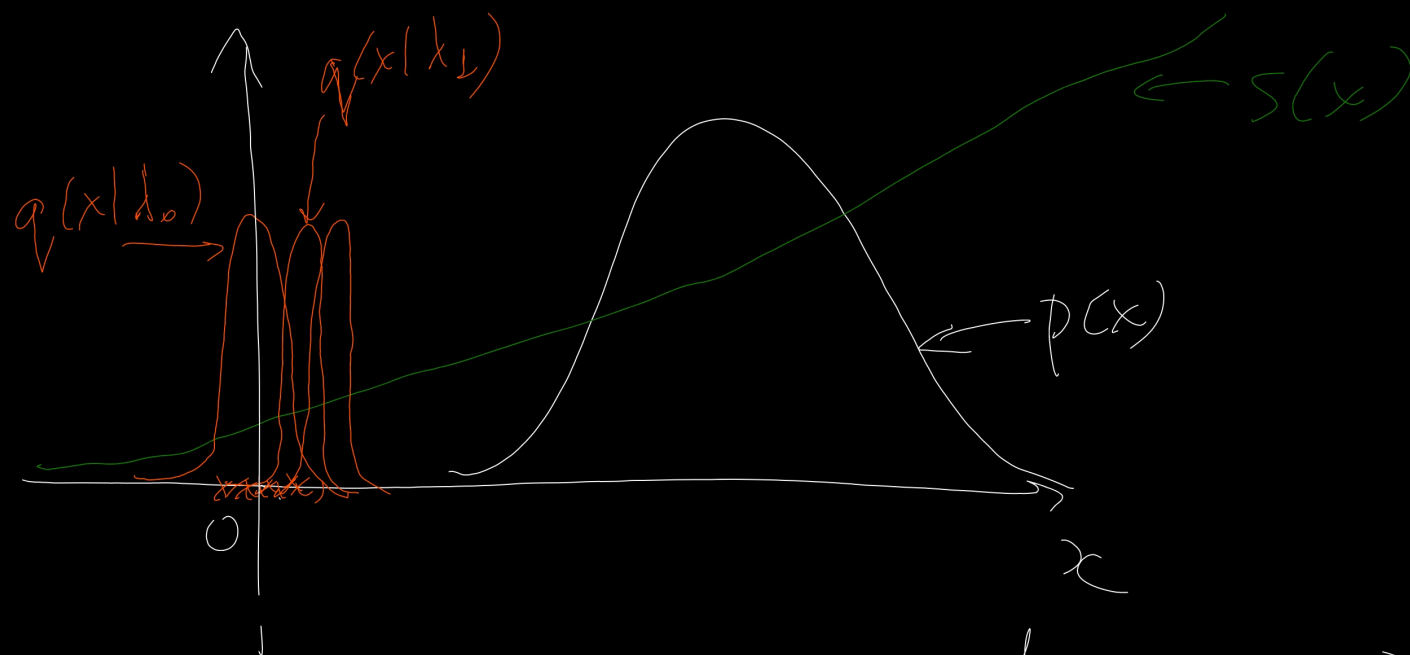
Step 1. $x_1, \dots, x_M \sim q(x | \lambda_t)$

Sort $S(x_{(1)}) \leq S(x_{(2)}) \leq \dots \leq S(x_{(M)})$

$\gamma_t = \min(\gamma, S_{\lfloor (1-p)M \rfloor})$

Step 2. $\lambda_{t+1} = \arg \max_{\lambda} \frac{1}{M} \sum_{j=1}^M [S(x_j) \geq \gamma_t] p(x_j) \log q(x_j | \lambda)$

Step 3. if $\gamma_t < \gamma$, $t = t+1$, go to step 1
otherwise exit



In practice:

- $\sigma_t^2 = \frac{1}{N} \sum_{j=1}^N (x_j - \mu)^2 + \epsilon_t$
- $\mu_{t+1} = \alpha \mu_{\text{new}} + (1 - \alpha) \mu_t$

CEM for opt.

$$S(x) \rightarrow \max_x$$

$$\mathbb{E}_{p(x)} [S(x) \geq \gamma_{\downarrow}]$$

$p(x)$ - uniform dist.

$$\gamma_{\downarrow} \approx \max_x S(x)$$

CEM for opt. scheme

Init. $M, p \in (0, 1), \lambda_0, t = 0$

Step 1 Sample $x_1, \dots, x_M \sim q(x|\lambda_t)$

$$S(x_{(1)}) \leq \dots \leq S(x_{(M)})$$

$$p(x) = q(x|\lambda_t)$$

$$r_t = S(x_{[(1-p)M]})$$

Step 2

$$\lambda_{t+1} = \arg \max_{\lambda} \frac{1}{M} \sum_{j=1}^M [S(x_j) \geq r_t] \cdot \log q(x_j|\lambda)$$

$t = t + 1$, go to step 1

$\forall a, s$

CEM for RL

$$E_{p(\tau|\theta)} R_0 \rightarrow \max_{\theta}$$

$$p(\tau|\theta) = p(s_0) \prod (a_t | s_t, \theta) \phi(s_{t+1} | s_t, a_t) \dots$$

$$s(x) = R_0(\tau=x)$$

$$\begin{aligned} \arg \max_{\theta} \quad & \frac{1}{M} \sum_{j=1}^M [s(x_j) \geq \tau_t] \log p(\tau_j | \theta) = \\ & = \frac{1}{M} \sum_j [\dots] \sum_{t=0}^{T_{\max}} \log \pi(a_t^j | s_t^j, \theta) \end{aligned}$$

$$\begin{aligned} \pi_{\text{opt}}(a|s) &= \\ &= \frac{\# [s_t^j = s, a_t^j = a] + 1}{\# [s_t^j = s] + |A|} \end{aligned}$$

CEM for Tetris

$$V(s|\omega) = \sum_i \omega_i \varphi_i(s)$$

$$\pi(a|s) = \arg \max_a V(s'(s,a)|\omega)$$

$\varphi_i(s)$:

- height of highest column
- diff. between heigh. columns
- number of holes
- number of full lines
- ...

$$\omega \sim N(\omega | \mu, \Sigma)$$